METRIC
A-A-59398A
October 18, 2016
SUPERSEDING
A-A-59398
January 09, 2004

COMMERCIAL ITEM DESCRIPTION

SHIELDING GASKET, ELECTRONIC

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers gasket material, principally comprised of a polymer gel substrate (silicone, polyurethane, or equivalent) used to displace moisture and provide environmental protection. Additionally, the substrate shall surround a woven matrix or deposited filler constructed from Monel®, aluminum, or other highly conductive metal that, when exposed during compression, provides electrical bonding for joints between surfaces for installations in shipboard and other exposed applications. The gaskets are applied to joints for which the faying surfaces have been prepared by cleaning to "bright and shiny" base metal.

2. CLASSIFICATION. The gaskets shall be of the following types, classes, and sizes:

2.1 Types and classes.

a. Format:
   1. Type I – Sheet gasket material (field-cut to size).
   2. Type II – Strip or cord gasket (field-cut to length), high-performance polymer gel, Monel®, aluminum, or equivalent kitted rope gasket.
   3. Type III – Pre-cut or pre-formed gasket.

b. Application:
   1. Class 1 – Standard polymer gel gasket.
   2. Class 2 – Polymer gel gasket, high salt spray resistance.
   3. Class 3 – Polymer gel gasket, high fuel resistance.

2.2 Sizes. Gasket material shall be specified by citation of the following dimensions for each type:

a. Type I – XX mm (XX in) length by YY mm (YY in) width by ZZ mm (ZZ in) thick.
b. Type II – XX mm (XX in) diameter by YY mm (YY in) spool length.
c. Type III – Custom cut in accordance with user's requirements.

3. SALIENT CHARACTERISTICS.

3.1 Materials. The gasket shall be fabricated from a metallic matrix, encapsulated within a pliable, non-hardening gel. The gel provides a moisture-repelling environmental seal; the metallic matrix provides electrical bonding when the gasket material is adequately compressed, thereby exposing the metallic matrix to the faying surfaces being joined.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:13Z
Check the source to verify that this is the current version before use.

3.2 Operating environment. The gaskets are intended for use with equipment installations in exposed, unsheltered environments aboard surface ships and in exposed environments in shore and land-mobile applications. The expected service life within this operating environment shall be for the service life of the joint. Using these gaskets in less exposed or interior locations may also be appropriate where similar environmental protection or electrical conduction requirements exist.

3.3 Performance. This description addresses the construct and performance of the gasket material only and does not intend to address application methodology or procedures. Where electromagnetic interference (EMI) protection is required, installers should consult installation design, equipment manufacturer's technical manuals, and/or prevailing topside EMI mitigation guidance for gasket usage, location, positioning, and coverage requirements.

The gaskets shall have the following performance characteristics:

a. Tack: adhere to prepared surfaces with enough adhesion to hold the weight of the gasket in any orientation, but not so much as to cause destruction of the gasket if repositioned.

b. Non-hardening: be removable for re-entry or opening of the sealed joint. NOTE: The gaskets covered by this CID are neither intended nor desired to be used more than once.

c. Non-toxic: not require special handling in accordance with Occupational Safety and Health Administration (OSHA) guidelines.

d. Conductive: provide direct current electrical resistance of less than 2.5 milliohms across the mechanical joint in accordance with MIL-STD-464 when fully installed.

e. Require no cure cycle or air-tight handling prior to use.

f. Temperature tolerant: remain fully functional (non-hardening/non-melting) throughout a -25 to 125 °C temperature range when tested in accordance with MIL-STD-810.

g. For Class 2 and 3 gaskets: seal, waterproof, and provide bonding for bimetallic joints (provide protection against galvanic corrosion) when tested with external performance requirement testing in accordance with ASTM B117.

h. For Class 3 gaskets: provide resistance to fuels when tested in accordance with MIL-STD-810, Method 504.2, Contamination By Fluids.

4. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. PRODUCT CONFORMANCE PROVISIONS. The products provided shall meet the salient characteristics of this commercial item description; conform to the producer's own drawings, specifications, standards, and quality assurance practices; and be functionally equivalent to the same product offered for sale in the commercial marketplace. The Government reserves the right to require proof of such conformance.

6. PACKAGING.

6.1 Packaging. Preservation, packing, and marking shall be as specified in the contract or order.

6.2 Labeling. The kits shall be labeled as to dimensions, or when appropriate, showing the antenna model/series that the kit is designed to accommodate.

6.3 Bulk packaging. Gaskets provided in bulk shall be packed so as to provide protection for gaskets remaining in the package. Each package shall include the gasket material and an instruction sheet.

7. NOTES.

7.1 Source of documents.

7.1.1 ASTM International. ASTM standards are available from ASTM International, 100 Barr Harbor Dr., P.O. Box C700, West Conshohocken, PA 19428-2959 or online at http://www.astm.org.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:13Z
Check the source to verify that this is the current version before use.

7.1.2 Defense specifications and standards. Defense specifications and standards are available from the Standardization Document Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094 or online at http://quicksearch.dla.mil.

7.1.3 FAR. The Federal Acquisition Regulation may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402 or online at https://www.acquisition.gov/far.

7.1.4 Occupational Safety and Health Administration (OSHA). OSHA guidelines are available from the U.S. Department of Labor/OSHA, OSHA Publications, P.O. Box 37535, Washington, DC 20013-7535 or online at https://www.osha.gov.

7.2 National stock numbers (NSNs). NSNs for gaskets can be found within the Federal Logistics Information Service (FLIS) database (FedLog) by using the manufacturer's name or their Commercial and Government Entity (CAGE) code. Manufacturers known to the Government that produce conductive gaskets are listed below. This list is not all-inclusive, and listing a manufacturer herein does not constitute endorsement by the Government.

a. Aviation Devices and Electronics Components, LLC (Av-DEC), CAGE 1NPE1.

b. Parker-Hannifin Corporation, Chomerics Division, CAGE U4299.

c. 3M Company, Industrial Adhesives and Tapes Division, CAGE 52152.

7.3 Shelf life. The shelf life of the gasket material shall be not less than 60 months when stored in a sheltered space having a controlled environment.

7.4 Ordering data. The contract or order should specify the following:

a. CID document number and revision.

b. Product conformance provisions.

c. Packaging requirements.

d. Quantity, type, class, and size required.

7.5 Key words.

Aluminum matrix
Antenna gaskets
Conductive gaskets
Electrical bonding
Gasket material
Monel® matrix
Polymer gel
Polyurethane gel
Silicone gel
Weatherproofing

7.6 Standard sizes. Gaskets that are custom cut to accommodate specific antenna models are available from multiple manufacturers as shown in 7.2.

MILITARY INTERESTS:

Custodian:
Navy – SH

Review Activity:
DLA – CC

CIVIL AGENCY COORDINATING ACTIVITY:
GSA – FAS

Preparing Activity:
Navy – SH
(Project 2090-2016-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:13Z
Check the source to verify that this is the current version before use.